

AMS 580 Logistic Regression HW

Kachun Huang

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Logistic Regression (* A type of Generalized Linear Model) with the Banknote Data The banknote.csv data (see attached) were extracted from images that were taken from genuine and forged banknote-like specimens. Yes, this is a Catch Me if You Can story. For digitization, an industrial camera usually used for print inspection was used. The final images have 400x 400 pixels. Wavelet Transform tool were used to extract features from images. There are 1372 banknotes, and 5 variables:

1. variance of Wavelet Transformed image (continuous)
2. skewness of Wavelet Transformed image (continuous)
3. curtosis of Wavelet Transformed image (continuous)
4. entropy of image (continuous)
5. class (binary) – this is the response variable of interest, 1 (forged) or 0 (genuine).

Import library

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.0
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.3.3
```

```
## Loading required package: lattice
```

```
##
```

```
## Attaching package: 'caret'
```

```
##
## The following object is masked from 'package:purrr':
##
## lift
```

```
library(MASS)
```

```
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
## select
```

Data Import

```
banknote <- read.csv("/Users/thomashuang/Downloads/banknote.csv")
str(banknote)
```

```
## 'data.frame': 1372 obs. of 5 variables:
## $ variance: num 3.622 4.546 3.866 3.457 0.329 ...
## $ skewness: num 8.67 8.17 -2.64 9.52 -4.46 ...
## $ curtosis: num -2.81 -2.46 1.92 -4.01 4.57 ...
## $ entropy : num -0.447 -1.462 0.106 -3.594 -0.989 ...
## $ class : int 1 1 1 1 1 1 1 1 1 1 ...
```

Data pre-processing

Handle Missing Value

```
banknote <- na.omit(banknote)
```

Factor

Since class should be a factor type not int type, we need to convert from int to factor .

```
banknote$class <- factor(banknote$class, levels = c(0,1))
str(banknote)
```

```
## 'data.frame': 1372 obs. of 5 variables:
## $ variance: num 3.622 4.546 3.866 3.457 0.329 ...
## $ skewness: num 8.67 8.17 -2.64 9.52 -4.46 ...
## $ curtosis: num -2.81 -2.46 1.92 -4.01 4.57 ...
## $ entropy : num -0.447 -1.462 0.106 -3.594 -0.989 ...
## $ class : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
```

Data Splitting

Please split the data into 80% training and 20% testing using seed =123.

```
set.seed (123)
training.samples <- banknote$class %>% createDataPartition(p=0.8, list=FALSE)
train.data <- banknote[training.samples,]
test.data <- banknote[-training.samples,]
```

Model Fitting

Then you shall fit a logistic regression model with all 4 predictors using the training data.

```
full.model <- glm(class ~., family = binomial, data = train.data)
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
summary(full.model)
```

```
##
## Call:
## glm(formula = class ~ ., family = binomial, data = train.data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -8.5895     2.1863  -3.929 8.54e-05 ***
## variance      9.3611     2.4703   3.789 0.000151 ***
## skewness      4.6968     1.2673   3.706 0.000210 ***
## curtosis      6.1372     1.6414   3.739 0.000185 ***
## entropy       0.5193     0.4208   1.234 0.217156
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1508.568  on 1097  degrees of freedom
## Residual deviance:   33.991  on 1093  degrees of freedom
## AIC: 43.991
##
## Number of Fisher Scoring iterations: 13
```

Prediction

- (c) Please use this fitted model based on the training data to predict the response variable 'class' (whether the banknote is forged or not) for the testing data.

```
pred_prob <- full.model %>% predict (test.data, type="response")
predicted.classes <- ifelse(pred_prob > 0.5, 1, 0)
```

Confusion Matrix

Please generate the confusion matrix, and report: (i) The overall accuracy;

```
confu_matrix <- confusionMatrix(factor(predicted.classes), factor(test.data$class), positive = "1")
confu_matrix
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  0    1
##           0 119    2
##           1   3 150
##
##           Accuracy : 0.9818
##           95% CI : (0.9579, 0.994)
##       No Information Rate : 0.5547
##       P-Value [Acc > NIR] : <2e-16
##
##           Kappa : 0.963
##
##  Mcnemar's Test P-Value : 1
##
##           Sensitivity : 0.9868
##           Specificity : 0.9754
##           Pos Pred Value : 0.9804
##           Neg Pred Value : 0.9835
##           Prevalence : 0.5547
##           Detection Rate : 0.5474
##       Detection Prevalence : 0.5584
##           Balanced Accuracy : 0.9811
##
##           'Positive' Class : 1
##
```

When positive class = “1” Detect forged

Actual/Predicted	Negative (Actual 0)	Positive (Actual 1)
Negative (Predicted 0)	True Negative (TN)	False Negative(FN)
Positive (Predicted 1)	False Positive(FP)	True Positive(TP)

Sensitivity

(ii) The sensitivity (that is, the probability a banknote is predicted to be forged given that it was in fact forged);

```
confu_matrix$byClass["Sensitivity"]
```

```
## Sensitivity
## 0.9868421
```

Specificity

- (iii) The specificity (that is, the probability a banknote is predicted to be genuine given that it was in fact genuine).

```
confu_matrix$byClass["Specificity"]
```

```
## Specificity  
## 0.9754098
```

Accuracy

```
confu_matrix$overall["Accuracy"]
```

```
## Accuracy  
## 0.9817518
```